

# 吴筱琦

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现居城市: 伦敦/苏州

意向城市: 上海/苏州/杭州 | 生态与环境数据分析



## 教育背景

帝国理工学院 (QS 2) 计算生命科学研究型硕士 伦敦 2025.09 - 2026.09

- 课程内容: 生态学与进化生物学中的定量方法, 包括数据科学、GIS、基因组学与生物信息学

帝国理工学院 (QS 2) 生物科学本科 伦敦 2022.09 - 2025.07

- 相关课程: 生态学与进化论, 生态学基础, 行为生态学, 生态学实地技能, 全球变化生物学, 生物多样性与保护生物学, 编程与统计

## 实习与社会实践

上海宣泰医药科技有限公司 分析工程师 · 分析部 上海 2024.07 - 2024.09

- 数据分析: 独立完成样本前处理、HPLC 系统操作及数据分析, 在严格的质控流程下确保检测结果的准确性和可靠性, 积累了跨领域量化数据分析经验

## 科研与项目经历

多模态大语言模型提取媒介性状数据评估: 高精确率与结构依赖型记录召回 伦敦 2025.12 - 2026.08

Python、Gemma 4 31B IT (多模态 LLM)、PyMuPDF、Docling、提示工程、JSON Schema 校验、pytest

- 基准构建与提取流程: 设计加权多样性贪心算法从 VecTraits 超 4 万行、334 篇论文中选取 20 篇, 构建 2,335 条开发基准与 431 条留出测试集; 结合 PyMuPDF/Docling 证据提取驱动 Gemma 4 31B IT 完成性状记录的多模态提取, 为生态学文献数据整理提供可复用的量化质控框架
- 严谨评估: 设计二分图记录匹配评估框架, 在留出测试集上验证锁定流程, 取得记录精确率 1.000、条件宏观字段 F1 0.729 (开发集经 11 轮提示词迭代达 0.810)

气候变化对佛罗里达州埃及伊蚊种群动态的影响建模 伦敦 2025.03 - 2025.06

R、deSolve、ggplot2、ODEs、统计验证 (RMSE、Pearson、Spearman)

- 模型适配与验证: 将已发表的热带 ODE 房室模型适配用于模拟佛罗里达 Collier 县埃及伊蚊种群在温度/降雨影响下的动态变化; 基于 53 周监测数据验证模型 (RMSE=1.65, Pearson  $r=0.29$ , Spearman  $\rho=0.30$ )
- 研究成果: 优化时间滞后 ( $\tau = -2$  周) 与降雨尺度因子, 提出模型改进方向 (人为水源、空间异质性、不确定性量化)

蜂鸟宏观生态学分析: 纬度梯度与生态学规则验证 伦敦 2024.10 - 2024.12

R、ggplot2、caper、GLM、PGLS

- 假设检验: 基于 AVONET 数据与系统发育树, 应用 GLM 与 PGLS 检验纬度多样性梯度 (LDG)、伯格曼法则和拉波波特法则; 结果强力支持 LDG, 且伯格曼法则在控制系统发育信号 (PGLS) 后由显著转为不显著, 体现了区分统计相关性 with 系统发育混杂因素的方法论严谨性

环境花色比例对昆虫颜色选择的影响 开普敦 2024.09 - 2024.10

野外实验、R (t 检验、Pearson 相关)

- 实验设计: 在南非野外课程中, 于 6 个样地设计五色陷阱盘实验, 研究昆虫访花颜色偏好与环境花色比例的关系
- 实验结果: 黄色、蓝色和白色陷阱均显著高于基线吸引昆虫 ( $p<0.001$ ); 黄色捕获率与环境黄花比例显著正相关 ( $r=0.369$ ,  $p=0.027$ )

## 技能/其他

- 数据分析与建模: R (ggplot2、dplyr、caper、ape、vegan、deSolve)、Python、统计推断 (GLM/泊松回归、PGLS、Wilcoxon 检验、t 检验、Pearson/Spearman 相关)、基于微分方程与差分方程的种群动力学建模
- 生态与野外研究: 野外实验设计、生物多样性/环境调查、基于 GIS 的空间分析、宏观生态学与系统发育比较分析 (AVONET 级性状数据、PGLS)
- 语言: 英语 (雅思 7.5), 普通话 (母语), 法语 (A2); 兴趣爱好: 钢琴, 摄影, 滑雪

# Xiaoqi Wu

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Currently based in: London / Suzhou

Ecological & Environmental Data Analysis | Shanghai / Suzhou / Hangzhou

## Education

**Imperial College London** MSc Computational Methods in Ecology and Evolution    London 2025.09 - 2026.09

- **Curriculum:** Quantitative methods in ecology and evolutionary biology, including data science, **GIS**, genomics, and bioinformatics

**Imperial College London** BSc Biological Sciences    London 2022.09 - 2025.07

- **Relevant Coursework:** Ecology and Evolution, Fundamentals of Ecology, Behavioural Ecology, Field Skills in Ecology, Global Change Biology, Biodiversity and Conservation Biology, Programming and Statistics

## Internship & Engagement

**Shanghai Sunherb Pharmaceutical Technology Co., Ltd.** Analytical Engineer · Analytical Department    Shanghai 2024.07 - 2024.09

- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis under strict quality-control procedures, ensuring accuracy and reliability – building cross-domain quantitative data-analysis experience

## Research & Project Experience

**Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery**    London 2025.12 - 2026.08

*Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest*

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard – a reusable quantitative QC framework applicable to ecological data curation
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Results:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics); evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000 and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile suited to human-in-the-loop curation

**Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida**    London 2025.03 - 2025.06

*R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)*

- **Model Adaptation & Validation:** Adapted a published tropical ODE compartment model to simulate *Aedes aegypti* population dynamics in Collier County, Florida under temperature/rainfall effects; validated against 53 weeks of surveillance data (RMSE=1.65, Pearson  $r=0.29$ , Spearman  $\rho=0.30$ )
- **Outcome:** Optimized time-lag ( $\tau = -2$  weeks) and a rainfall scaling factor; proposed model improvements (anthropogenic water sources, spatial heterogeneity, uncertainty quantification)

**Macroecology Analysis on Trochilidae: Testing Latitudinal Gradients and Ecological Rules**    London 2024.10 - 2024.12

*R, ggplot2, caper, GLM, PGLS*

- **Hypothesis Testing:** Tested the latitudinal diversity gradient (LDG), Bergmann's rule, and Rapoport's rule using AVONET data and a phylogenetic tree, applying GLM and PGLS to model species richness against body mass, range size, and latitude
- **Findings:** Results strongly supported the LDG; Bergmann's rule was significant under a naive GLM but lost significance once phylogenetic signal was controlled for (PGLS) – a result that demonstrates the methodological rigor of separating statistical correlation from phylogenetic confounding

### **Effect of Environmental Flower Colour Ratios on Insect Colour Preference** Cape Town 2024.09 - 2024.10

*Field experiment, R (t-test, Pearson correlation)*

- **Experimental Design:** Designed a 5-colour pan-trap field experiment across 6 sites during a South Africa field course to study insect flower-visiting colour preferences relative to environmental flower colour ratios
- **Results:** Yellow, blue, and white traps all significantly attracted insects over baseline ( $p < 0.001$ ); yellow capture rate was positively correlated with the proportion of yellow flowers in the environment ( $r = 0.369$ ,  $p = 0.027$ )

### **Skills & Other**

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- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, ape, vegan, deSolve), Python, statistical inference (GLM/Poisson regression, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE- and difference-equation-based population dynamics modelling
- **Ecology & Fieldwork:** field experimental design, biodiversity/environmental surveying, GIS-based spatial analysis, macroecology and phylogenetic comparative analysis (AVONET-scale trait data, PGLS)
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, photography, skiing